

ERP SEKOLAH ALAM

Blueprint Architecture Document — Level 1 sampai Level 9

Versi: 2.0 (Revisi Komprehensif) **Tanggal:** Juni 2026 **Penyusun:** ERP Architect **Stack**
Target: Native PHP 8.2 · MySQL 8 · Bootstrap 5 · No Framework · No API

RINGKASAN EKSEKUTIF

ERP Sekolah Alam **bukan** Student Information System (SIS) konvensional. Ini adalah **Outcome-Based Daily Learning Management System** yang dibangun di atas siklus operasional:

Plan → Do → Observe → Assess → Report → Portfolio → Progress → Semester Report

Jantung operasional sistem bukan pada nilai atau jadwal mata pelajaran, melainkan pada:

- Daily Plan (rencana kegiatan harian fasilitator)
- Daily Execution & Observation (pelaksanaan & pencatatan LOA)
- Daily Report (laporan harian ke orang tua)
- Portfolio (kumpulan evidence pembelajaran)
- Progress & Semester Report (rapor berbasis evidence + outcome)

LEVEL 1 — BUSINESS DISCOVERY

1.1 Visi Sistem

Menyediakan platform terpadu yang mengelola seluruh siklus pendidikan Sekolah Alam mulai dari penerimaan siswa baru, perencanaan pembelajaran harian, pelaksanaan kegiatan, pengukuran Learning Outcome, dokumentasi portofolio siswa, pelaporan harian kepada orang tua, hingga pelaporan manajemen dan keuangan.

1.2 Business Goal

Goal ID	Business Goal
BG-001	Digitalisasi proses PPDB (Penerimaan Peserta Didik Baru)
BG-002	Sentralisasi data siswa, guru, dan orang tua
BG-003	Mendukung pembelajaran berbasis alam dan pengalaman (experiential learning)

Goal ID Business Goal

BG-004	Mendukung pengukuran Learning Outcome Assessment (LOA) berbasis observasi harian
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BG-005	Menyediakan komunikasi transparan kepada orang tua melalui Daily Report
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BG-006	Mempermudah pengelolaan event sekolah (Market Day, Outbound, Wisuda, dll.)
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BG-007	Mengelola tagihan pendidikan secara terstruktur
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BG-008	Menyediakan dashboard dan laporan manajemen untuk pengambilan keputusan
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BG-009	Menyediakan sistem dokumentasi pembelajaran harian yang terstruktur
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BG-010	Mengubah seluruh aktivitas pembelajaran menjadi evidence pembelajaran
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BG-011	Menghasilkan portofolio siswa secara otomatis dari evidence harian
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BG-012	Menghasilkan rapor semester berbasis evidence dan outcome (OBE)
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1.3 Stakeholder Catalog Matrix

Stakeholder	Stakeholder Type	Tujuan Utama	Kebutuhan Utama
Yayasan / Owner	Sponsor	Pertumbuhan & keberlanjutan institusi	KPI sekolah, laporan keuangan, dashboard eksekutif
Kepala Sekolah	Decision Maker	Pengendalian mutu & pengambilan keputusan	Monitoring akademik, approval, laporan performa
Wakil Kepala Sekolah	Decision Maker	Koordinasi program & kurikulum	Monitoring kelas, jadwal, capaian LOA
Admin Sekolah	Administrator	Tata kelola data & operasional sistem	PPDB, data siswa/guru, tagihan, user management
Guru / Fasilitator	Operational User	Melaksanakan & mendokumentasikan pembelajaran	Daily Plan, observasi LOA, daily report, portofolio
Wali Kelas	Operational User	Mengelola kelas & komunikasi orang tua	Absensi, rapor, progress siswa, daily report kelas
Siswa	Internal User	Mengakses materi & melihat perkembangan	Jadwal, tugas, nilai, portofolio pribadi

Stakeholder	Stakeholder Type	Tujuan Utama	Kebutuhan Utama
Orang Tua / Wali Murid	External Party	Memantau perkembangan & memenuhi kewajiban	Daily report, LOA anak, tagihan SPP, event
Staff Keuangan	Operational User	Pengelolaan keuangan sekolah	Tagihan, pembayaran, verifikasi, laporan keuangan
Panitia Event	Operational User	Mengelola event & kegiatan sekolah	Perencanaan event, registrasi peserta, evaluasi
Auditor Internal	Support	Menjaga kepatuhan & audit trail	Audit log, perubahan data, kepatuhan bisnis rules

1.4 Business Capability Map



- └ Daily Activity Recording
- └ Daily Observation (LOA)
- └ Daily Assessment
 - └ Daily Reporting

LOA (LEARNING OUTCOME ASSESSMENT) DOMAIN

- └ Outcome Definition
- └ Observation Management
- └ Evidence Collection
- └ Outcome Evaluation
 - └ LOA Reporting

STUDENT PORTFOLIO DOMAIN ← Domain Baru

- └ Portfolio Collection (dari evidence harian)
- └ Portfolio Compilation
 - └ Portfolio Publishing (ke orang tua)

STUDENT PROGRESS DOMAIN ← Domain Baru

- └ Progress Monitoring
- └ Progress Reporting
 - └ Parent Communication

ACADEMIC MANAGEMENT DOMAIN

- └ Curriculum Management
- └ Subject Management
- └ Class Management
- └ Schedule Management
- └ Attendance Management
- └ Assessment Management
 - └ Report Card Management (OBE-based)

LMS DOMAIN

- └ Learning Material Management
 - └ Assignment Management

EVENT MANAGEMENT DOMAIN

- └ Event Planning
- └ Event Registration
- └ Event Execution
 - └ Event Evaluation

FINANCE MANAGEMENT DOMAIN

- └ Registration Billing
- └ Tuition Billing
- └ Payment Collection
 - └ Payment Verification

HUMAN RESOURCE DOMAIN

- └ Teacher Management
- └ Teacher Assignment
- └ Teacher Evaluation

REPORTING & ANALYTICS DOMAIN

- └ Executive Dashboard
- └ Academic Dashboard
- └ LOA Dashboard
- └ Finance Dashboard
- └ Event Dashboard

SYSTEM DOMAIN

- └ User Management
- └ Role & Permission
- └ Audit Trail

LEVEL 2 — BUSINESS ARCHITECTURE

2.1 Business Resource Catalog

Master Resource

Resource	Deskripsi
Student	Data siswa aktif, alumni
Parent	Data orang tua / wali murid
Teacher / Facilitator	Data guru & fasilitator
Class	Data kelas (nama kelas, tingkat, ruang)
AcademicYear	Tahun ajaran & semester
Subject / Mapel	Mata pelajaran / tema
LearningOutcome	Definisi outcome (LO-01 s/d LO-06)
Event	Data event sekolah
User	Akun pengguna sistem
Role	Peran pengguna

Planning Resource

Resource	Deskripsi
SemesterPlan	Rencana tema besar per bulan dalam satu semester
WeeklyPlan	Rencana aktivitas per minggu
DailyPlan	Rencana kegiatan harian (wizard 4-step)
Theme	Tema pembelajaran
LearningObjective	Tujuan pembelajaran spesifik per hari

Execution Resource

Resource	Deskripsi
DailyActivity	Aktivitas yang dilaksanakan dalam sehari
Attendance	Data kehadiran siswa per hari
Observation	Catatan observasi LOA per siswa per hari
DailyAssessment	Penilaian harian berbasis observasi
DailyReport	Laporan harian yang dikirim ke orang tua

Evidence Resource

Resource	Deskripsi
LearningEvidence	Evidence pembelajaran (foto, video, karya)
PhotoEvidence	Evidence foto kegiatan
VideoEvidence	Evidence video kegiatan
StudentWork	Karya/produk siswa

Outcome Resource

Resource	Deskripsi
PortfolioItem	Satu item portofolio dari evidence
Portfolio	Kumpulan portofolio siswa dalam satu periode
ProgressReport	Laporan progress bulanan / mingguan
SemesterReport	Rapor akhir semester berbasis OBE

Transaction Resource

Resource	Deskripsi
Applicant	Calon siswa baru (PPDB)
Registration	Data registrasi siswa
AssignmentSubmission	Pengumpulan tugas siswa
AcademicAssessment	Penilaian akademik
LOAObservation	Observasi LOA (linked ke Observation)
EventRegistration	Registrasi peserta event
Billing	Tagihan SPP / biaya lain
Payment	Pembayaran tagihan

System Resource

Resource	Deskripsi
AuditLog	Log seluruh perubahan data
EventLog	Log event bisnis
Notification	Notifikasi sistem ke pengguna

2.2 Business Activity Catalog

PPDB

- Create Applicant, Verify Applicant, Schedule Interview
- Conduct Assessment, Approve Applicant, Reject Applicant
- Register Student (convert Applicant → Student)

Learning Planning

- Create Semester Plan, Create Weekly Plan
- Create Daily Plan (wizard: Info Umum → Tujuan & Tema → Aktivitas → Materi & Penutup)
- Publish Daily Plan

Daily Execution

- Record Attendance, Execute Daily Activity
- Record Observation (LOA), Upload Evidence
- Input Daily Assessment, Create Daily Report
- Publish Daily Report (kirim ke orang tua)

Portfolio

- Collect Evidence → Create Portfolio Item
- Compile Portfolio, Publish Portfolio

Progress & Reporting

- Generate Weekly Summary, Generate Monthly Progress
- Generate Semester Report, Publish Semester Report (Rapor)

Academic

- Create Schedule, Manage Class
- Create Assignment, Review Submission
- Generate Report Card

LOA

- Define Outcome, Create Observation
- Upload Evidence, Assess Outcome
- Publish LOA Report

Event

- Create Event, Publish Event
- Open Registration, Register Participant
- Execute Event, Evaluate Event, Publish Report

Finance

- Generate Billing, Send Billing Notification

- Verify Payment, Approve Payment, Print Receipt

2.3 Draft ERP Navigation Blueprint

Dashboard ← role-based dashboard

Master Data ← Administrator only

- └ Siswa
 - └ Daftar Siswa
 - └ Tambah Siswa
 - └ Import Siswa (CSV)
- └ Orang Tua / Wali Murid
 - └ Daftar Orang Tua
 - └ Hubungkan ke Siswa
- └ Guru / Fasilitator
 - └ Daftar Guru
 - └ Tambah Guru
- └ Kelas
 - └ Daftar Kelas
 - └ Penempatan Siswa
- └ Tahun Ajaran
 - └ Daftar Tahun Ajaran
 - └ Atur Tahun Aktif
- └ Mata Pelajaran / Tema
 - └ Daftar Mapel / Tema
 - └ Tambah Mapel / Tema
- └ Learning Outcome
 - └ Daftar Learning Outcome
 - └ Tambah / Edit Outcome

Learning Planning

- └ Semester Plan
- └ Weekly Plan
- └ Daily Plan (wizard form)

Learning Execution

- └ Attendance
- └ Daily Activity
- └ Observation (LOA)
- └ Daily Assessment

Daily Report

- └ Create Report
- └ Publish Report

Portfolio

- └ Portfolio Item (evidence)

- └ Student Portfolio

Progress

- └ Progress Report

Semester Report

- └ Semester Assessment

- └ Semester Report

- └ Rapor (OBE)

Students

- └ Student Profile

- └ Parent Profile

PPDB (Admissions)

- └ Applicant

- └ Selection & Interview

- └ Registration

Academic

- └ Schedule

- └ Class Management

- └ Assignment

- └ Assessment

LOA

- └ Outcome Definition

- └ Observation

- └ Evidence

- └ Evaluation

- └ LOA Report

Events

- └ Event Calendar

- └ Event Registration

- └ Event Evaluation

Finance

- └ Billing

- └ Payment

Reports

- └ Academic Reports

- └ LOA Reports

- └ Finance Reports

- └ Event Reports

System (Administrator only)

- └ User Management
- └ Role & Permission
- └ Audit Trail
- └ System Configuration

LEVEL 3 — WORKFLOW DISCOVERY

3.1 Cross-Domain Workflow Utama

PPDB → Registrasi → Menjadi Siswa → Penempatan Kelas
→ Semester Plan → Weekly Plan → Daily Plan
→ Daily Execution → Observation → LOA Assessment
→ Daily Report (→ Orang Tua)
→ Evidence → Portfolio Item → Portfolio
→ Progress Report → Semester Report → Rapor
→ Kenaikan Kelas → (ulang) / Kelulusan → Alumni

3.2 Workflow Detail

Workflow 1 — PPDB & Admission

Create Applicant → Verify Dokumen → Schedule Interview
→ Conduct Assessment → Decision (Approve/Reject)
→ [Approve] Register Student → Create User Account → Assign Class
→ [Reject] Send Notification

Workflow 2 — Academic Planning (siklus tahunan)

Create AcademicYear → Define LearningOutcomes → Create SemesterPlan
→ Create WeeklyPlan (per minggu)
→ Create DailyPlan (per hari, wizard 4-step)
→ Publish DailyPlan

Workflow 3 — Daily Learning Execution (siklus harian)

DailyPlan [Published]
→ Record Attendance
→ Execute DailyActivity (catatan pelaksanaan)
→ Record Observation (per siswa, per outcome)
→ Upload Evidence (foto/video/karya)
→ Input DailyAssessment

Workflow 4 — Daily Communication

DailyAssessment + Observation

- Create DailyReport
- Publish DailyReport → Notification ke Orang Tua
- Orang Tua dapat view di Parent Portal

Workflow 5 — Portfolio Building (otomatis)

Evidence (dari observasi harian)

- Auto-create PortfolioItem
- Compile Portfolio (per siswa per semester)
- Publish Portfolio

Workflow 6 — Progress Monitoring

DailyAssessment (akumulasi)

- Weekly Summary (otomatis)
- Monthly Progress Report
- Semester Progress

Workflow 7 — Semester Reporting (Rapor OBE)

DailyAssessment (akumulasi semester)

- + Portfolio (evidence terkumpul)
- + LOA Achievement (capaian per outcome)
- Semester Assessment
- Semester Report
- Rapor (dicetak / dikirim ke orang tua)

Workflow 8 — LOA Assessment

Observation [Created] → Evidence [Uploaded]

- Outcome Assessment (per outcome per siswa)
- Validation oleh Wali Kelas
- Publish LOA Report

Workflow 9 — Event

Create Event → Publish Event → Open Registration

- Register Participant → Execute Event (absensi)
- Collect Evaluation → Publish Event Report

Workflow 10 — Finance (SPP)

Generate Billing (per bulan/periode)
→ Send Notification ke Orang Tua
→ Parent Pay → Upload Bukti
→ Staff Keuangan Verify
→ Approve Payment → Status: PAID
→ Print / Send Receipt

3.3 Workflow Resource Matrix

Workflow	Input	Process	Output
Admission	Applicant	Seleksi & verifikasi	Student
Daily Planning	WeeklyPlan	Wizard 4-step	DailyPlan
Daily Execution	DailyPlan	Absensi + aktivitas + observasi	DailyActivity + Observation
Portfolio	Evidence	Kompilasi otomatis	Portfolio
Progress	DailyAssessment	Agregasi periodik	ProgressReport
Semester Report	Assessment + Portfolio + LOA	Generate rapor	SemesterReport
Event	EventPlan	Registrasi + pelaksanaan	EventReport
Finance	Student	Generate + verifikasi	Payment

LEVEL 4 — EVENT MODELING

4.1 Event Catalog

Admission Events

APPLICANT_CREATED, APPLICANT_VERIFIED, APPLICANT_INTERVIEWED
APPLICANT_APPROVED, APPLICANT_REJECTED, STUDENT_REGISTERED

Planning Events

SEMESTER_PLAN_CREATED, WEEKLY_PLAN_CREATED
DAILY_PLAN_CREATED, DAILY_PLAN_PUBLISHED

Learning Events

ATTENDANCE_RECORDED, ACTIVITY_EXECUTED
OBSERVATION_RECORDED, EVIDENCE_UPLOADED
DAILY_ASSESSMENT_COMPLETED

Reporting Events

DAILY_REPORT_CREATED, DAILY_REPORT_PUBLISHED
PROGRESS_REPORT_PUBLISHED, SEMESTER_REPORT_PUBLISHED

Portfolio Events

PORTFOLIO_ITEM_CREATED, PORTFOLIO_COMPILED, PORTFOLIO_PUBLISHED

LOA Events

OUTCOME_DEFINED, LOA_OBSERVATION_CREATED
LOA_EVIDENCE_UPLOADED, OUTCOME_ASSESSED, LOA_REPORT_PUBLISHED

Academic Events

ASSIGNMENT_CREATED, ASSIGNMENT_SUBMITTED, ASSESSMENT_COMPLETED
REPORT_CARD_PUBLISHED

Event Events

EVENT_CREATED, EVENT_PUBLISHED, EVENT_REGISTERED
EVENT_COMPLETED, EVENT_EVALUATED

Finance Events

BILL_GENERATED, PAYMENT_RECEIVED
PAYMENT_VERIFIED, RECEIPT_PRINTED

System Events

USER_LOGIN, USER_LOGOUT, USER_CREATED
DATA_CHANGED, PERMISSION_CHANGED

4.2 State Transition Matrix

Resource	States
Applicant	Draft → Submitted → Verified → Interviewed → Approved/Rejected → Registered
Student	Registered → Active → Promoted → Graduated → Alumni
DailyPlan	Draft → Published
DailyReport	Draft → Published
Observation	Draft → Submitted → Validated
Portfolio	Collecting → Compiled → Published
Billing	Draft → Issued → Partially Paid → Paid
Payment	Pending → Verified → Approved
Event	Draft → Published → Registration Open → Ongoing → Completed

4.3 Business Rule Catalog

Rule Code	Rule Description
ADM-001	Calon siswa wajib melengkapi seluruh dokumen sebelum diproses
ADM-002	Siswa hanya dapat diregistrasi setelah status Applicant = Approved
PLN-001	Daily Plan hanya dapat dipublish jika minimal 1 aktivitas telah ditambahkan
PLN-002	Daily Plan harus terhubung ke Weekly Plan yang aktif
EXE-001	Absensi hanya dapat diinput pada hari yang sama dengan DailyPlan
LOA-001	Observasi LOA wajib memiliki minimal 1 evidence sebelum dapat disubmit
LOA-002	Evidence harus berupa file valid (foto/video/dokumen)
RPT-001	Daily Report hanya dapat dipublish setelah absensi direcord
PRT-001	Portfolio Item dibuat otomatis saat evidence diupload & observation divalidasi
SEM-001	Semester Report hanya bisa dibuat jika minimal 80% daily assessment telah diinput
ACD-001	Tugas hanya dapat dinilai setelah dikumpulkan (submission exists)
FIN-001	Tagihan dianggap lunas setelah pembayaran diverifikasi & diapprove

Rule Code	Rule Description
FIN-002	Satu billing hanya memiliki satu payment yang approved
SYS-001	Setiap perubahan data sensitif wajib dicatat di audit_log
SYS-002	User hanya dapat mengakses menu sesuai role yang ditetapkan

LEVEL 5 — DATA ARCHITECTURE

5.1 Entity Classification

Master Tables

users, roles, permissions, role_permissions
students, parents, teachers
classes, academic_years, subjects
learning_outcomes, events

Planning Tables

semester_plans, weekly_plans, daily_plans
daily_plan_activities, daily_plan_materials
learning_themes, learning_objectives

Execution Tables

attendances, daily_activities
loa_observations, loa_evidences
daily_assessments, daily_reports

Transaction Tables

applicants, registrations
assignments, assignment_submissions
academic_assessments
event_registrations
billings, payments

Outcome Tables

System Tables

```
audit_logs, event_logs, notifications
```

5.2 Conceptual ERD



5.3 Tabel Prioritas Tinggi (Core Tables)

```
sql
```

```

-- =====
-- SYSTEM
-- =====

CREATE TABLE roles (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50) NOT NULL,           -- 'kepala_sekolah', 'admin', 'guru',
    label VARCHAR(100) NOT NULL,         -- 'Kepala Sekolah'
    stakeholder_type ENUM('sponsor', 'decision_maker', 'operational_user',
                           'administrator', 'internal_user', 'external_party', 'sup
    description TEXT,
    created_at DATETIME DEFAULT NOW()
);

CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(50) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,      -- password_hash()
    full_name VARCHAR(150) NOT NULL,
    email VARCHAR(150),
    phone VARCHAR(20),
    role_id INT NOT NULL,
    entity_type ENUM('student', 'parent', 'teacher', 'staff', 'admin', 'other'),
    entity_id INT,                       -- FK ke tabel spesifik sesuai entity
    is_active TINYINT(1) DEFAULT 1,
    last_login DATETIME,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (role_id) REFERENCES roles(id)
);

CREATE TABLE audit_logs (
    id BIGINT PRIMARY KEY AUTO_INCREMENT,
    table_name VARCHAR(100),
    record_id INT,
    action ENUM('CREATE', 'UPDATE', 'DELETE', 'APPROVE', 'REJECT', 'LOGIN', 'LOGOUT')
    before_value JSON,
    after_value JSON,
    user_id INT,
    ip_address VARCHAR(45),
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (user_id) REFERENCES users(id)
);

-- =====
-- MASTER DATA
-- =====

CREATE TABLE academic_years (

```

```

    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(20) NOT NULL,          -- '2025/2026'
    semester ENUM('Ganjil', 'Genap'),
    start_date DATE,
    end_date DATE,
    is_active TINYINT(1) DEFAULT 0
);

CREATE TABLE learning_outcomes (
    id INT PRIMARY KEY AUTO_INCREMENT,
    code VARCHAR(10) NOT NULL,          -- 'LO-01'
    name VARCHAR(150) NOT NULL,         -- 'Karakter Mandiri'
    description TEXT,
    academic_year_id INT,
    is_active TINYINT(1) DEFAULT 1,
    FOREIGN KEY (academic_year_id) REFERENCES academic_years(id)
);

CREATE TABLE classes (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50) NOT NULL,          -- 'SD 4 - Saturnus'
    level VARCHAR(20),                  -- 'SD 4'
    room VARCHAR(100),
    academic_year_id INT,
    teacher_id INT,                     -- wali kelas
    is_active TINYINT(1) DEFAULT 1,
    FOREIGN KEY (academic_year_id) REFERENCES academic_years(id)
);

CREATE TABLE students (
    id INT PRIMARY KEY AUTO_INCREMENT,
    nis VARCHAR(20) UNIQUE,
    full_name VARCHAR(150) NOT NULL,
    nickname VARCHAR(50),
    gender ENUM('L', 'P'),
    birth_place VARCHAR(100),
    birth_date DATE,
    address TEXT,
    class_id INT,
    entry_year YEAR,
    status ENUM('active', 'graduated', 'transferred', 'inactive') DEFAULT 'active'
    photo VARCHAR(255),
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (class_id) REFERENCES classes(id)
);

CREATE TABLE parents (

```

```

    id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT NOT NULL,
    full_name VARCHAR(150) NOT NULL,
    relationship ENUM('Ayah', 'Ibu', 'Wali'),
    phone VARCHAR(20),
    email VARCHAR(150),
    occupation VARCHAR(100),
    address TEXT,
    is_primary TINYINT(1) DEFAULT 0,
    FOREIGN KEY (student_id) REFERENCES students(id)
);

CREATE TABLE teachers (
    id INT PRIMARY KEY AUTO_INCREMENT,
    nip VARCHAR(30),
    full_name VARCHAR(150) NOT NULL,
    email VARCHAR(150),
    phone VARCHAR(20),
    subjects TEXT, -- JSON array mapel/bidang
    is_active TINYINT(1) DEFAULT 1,
    photo VARCHAR(255),
    created_at DATETIME DEFAULT NOW()
);

-- =====
-- LEARNING PLANNING
-- =====

CREATE TABLE semester_plans (
    id INT PRIMARY KEY AUTO_INCREMENT,
    academic_year_id INT NOT NULL,
    class_id INT NOT NULL,
    month TINYINT,
    big_theme VARCHAR(200),
    learning_outcome_id INT,
    description TEXT,
    status ENUM('draft', 'published') DEFAULT 'draft',
    created_by INT,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (academic_year_id) REFERENCES academic_years(id),
    FOREIGN KEY (class_id) REFERENCES classes(id),
    FOREIGN KEY (learning_outcome_id) REFERENCES learning_outcomes(id)
);

CREATE TABLE weekly_plans (
    id INT PRIMARY KEY AUTO_INCREMENT,
    semester_plan_id INT NOT NULL,
    week_number TINYINT,

```

```

week_start DATE,
week_end DATE,
theme VARCHAR(200),
focus_activity TEXT,
status ENUM('draft','published') DEFAULT 'draft',
created_by INT,
created_at DATETIME DEFAULT NOW(),
FOREIGN KEY (semester_plan_id) REFERENCES semester_plans(id)
);

CREATE TABLE daily_plans (
  id INT PRIMARY KEY AUTO_INCREMENT,
  weekly_plan_id INT,
  class_id INT NOT NULL,
  plan_date DATE NOT NULL,
  day_name VARCHAR(15),
  theme VARCHAR(200) NOT NULL,
  sub_theme VARCHAR(200),
  learning_objective TEXT,
  quran_reference TEXT,           -- Referensi ayat Al-Quran
  akhlaq_focus TEXT,             -- Fokus karakter/akhlak (JSON array)
  opening_activity TEXT,         -- Kegiatan pembuka
  closing_activity TEXT,         -- Kegiatan penutup
  notes TEXT,
  status ENUM('draft','published') DEFAULT 'draft',
  created_by INT,
  published_at DATETIME,
  created_at DATETIME DEFAULT NOW(),
  FOREIGN KEY (weekly_plan_id) REFERENCES weekly_plans(id),
  FOREIGN KEY (class_id) REFERENCES classes(id)
);

CREATE TABLE daily_plan_activities (
  id INT PRIMARY KEY AUTO_INCREMENT,
  daily_plan_id INT NOT NULL,
  sort_order TINYINT DEFAULT 0,
  time_start TIME,
  time_end TIME,
  activity_title VARCHAR(200) NOT NULL,
  activity_type ENUM('outdoor','indoor','diskusi','kreasi','presentasi',
                    'ibadah','istirahat','lainnya') DEFAULT 'lainnya',
  description TEXT,
  tools_materials TEXT,           -- Alat & bahan
  learning_outcome_id INT,        -- LOA yang dituju
  FOREIGN KEY (daily_plan_id) REFERENCES daily_plans(id),
  FOREIGN KEY (learning_outcome_id) REFERENCES learning_outcomes(id)
);

```

```

-- =====
-- DAILY EXECUTION
-- =====

CREATE TABLE attendances (
    id INT PRIMARY KEY AUTO_INCREMENT,
    daily_plan_id INT NOT NULL,
    student_id INT NOT NULL,
    status ENUM('hadir', 'sakit', 'izin', 'alpha') DEFAULT 'hadir',
    notes TEXT,
    recorded_by INT,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (daily_plan_id) REFERENCES daily_plans(id),
    FOREIGN KEY (student_id) REFERENCES students(id)
);

CREATE TABLE loa_observations (
    id INT PRIMARY KEY AUTO_INCREMENT,
    daily_plan_id INT NOT NULL,
    student_id INT NOT NULL,
    learning_outcome_id INT NOT NULL,
    observation_text TEXT NOT NULL,
    achievement_level ENUM('berkembang', 'sesuai_harapan', 'melampaui'),
    score TINYINT, -- 1-100
    notes TEXT,
    status ENUM('draft', 'submitted', 'validated') DEFAULT 'draft',
    created_by INT,
    validated_by INT,
    validated_at DATETIME,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (daily_plan_id) REFERENCES daily_plans(id),
    FOREIGN KEY (student_id) REFERENCES students(id),
    FOREIGN KEY (learning_outcome_id) REFERENCES learning_outcomes(id)
);

CREATE TABLE loa_evidences (
    id INT PRIMARY KEY AUTO_INCREMENT,
    observation_id INT NOT NULL,
    file_type ENUM('photo', 'video', 'document', 'audio', 'other'),
    file_path VARCHAR(500) NOT NULL,
    file_name VARCHAR(255),
    file_size INT,
    caption TEXT,
    uploaded_by INT,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (observation_id) REFERENCES loa_observations(id)
);

```

```

CREATE TABLE daily_reports (
    id INT PRIMARY KEY AUTO_INCREMENT,
    daily_plan_id INT NOT NULL,
    class_id INT NOT NULL,
    report_date DATE NOT NULL,
    activity_summary TEXT,
    achievement_highlight TEXT,
    tomorrow_plan TEXT,
    overall_note TEXT,
    status ENUM('draft','published') DEFAULT 'draft',
    published_at DATETIME,
    created_by INT,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (daily_plan_id) REFERENCES daily_plans(id),
    FOREIGN KEY (class_id) REFERENCES classes(id)
);

-- =====
-- PORTFOLIO & PROGRESS
-- =====

CREATE TABLE portfolio_items (
    id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT NOT NULL,
    observation_id INT,
    evidence_id INT,
    title VARCHAR(200),
    item_type ENUM('karya','tulisan','video','foto','laporan','rekaman','lainny
description TEXT,
date_created DATE,
semester VARCHAR(20),
academic_year_id INT,
is_featured TINYINT(1) DEFAULT 0,
created_at DATETIME DEFAULT NOW(),
FOREIGN KEY (student_id) REFERENCES students(id)
);

CREATE TABLE semester_reports (
    id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT NOT NULL,
    class_id INT NOT NULL,
    academic_year_id INT NOT NULL,
    attendance_present TINYINT,
    attendance_sick TINYINT,
    attendance_permission TINYINT,
    attendance_alpha TINYINT,
    overall_grade VARCHAR(5),
    -- A, B+, B, dll

```

```

overall_score DECIMAL(5,2),
loa_summary JSON,                                -- {LO-01: 82, LO-02: 75, ...}
homeroom_notes TEXT,
principal_notes TEXT,
status ENUM('draft','published') DEFAULT 'draft',
published_at DATETIME,
created_at DATETIME DEFAULT NOW(),
FOREIGN KEY (student_id) REFERENCES students(id),
FOREIGN KEY (class_id) REFERENCES classes(id)
);

-- =====
-- PPDB & ADMISSIONS
-- =====

CREATE TABLE applicants (
    id INT PRIMARY KEY AUTO_INCREMENT,
    registration_number VARCHAR(30) UNIQUE,
    full_name VARCHAR(150) NOT NULL,
    birth_place VARCHAR(100),
    birth_date DATE,
    gender ENUM('L','P'),
    parent_name VARCHAR(150),
    parent_phone VARCHAR(20),
    parent_email VARCHAR(150),
    address TEXT,
    target_class VARCHAR(20),
    academic_year_id INT,
    interview_date DATE,
    interview_notes TEXT,
    assessment_score DECIMAL(5,2),
    status ENUM('draft','submitted','verified','interviewed',
               'approved','rejected','registered') DEFAULT 'draft',
    decision_notes TEXT,
    processed_by INT,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (academic_year_id) REFERENCES academic_years(id)
);

-- =====
-- FINANCE
-- =====

CREATE TABLE billings (
    id INT PRIMARY KEY AUTO_INCREMENT,
    billing_number VARCHAR(30) UNIQUE,
    student_id INT NOT NULL,
    billing_type ENUM('spp','daftar_ulang','kegiatan','lainnya') DEFAULT 'spp',
    description VARCHAR(200),

```

```

    amount DECIMAL(12,2) NOT NULL,
    due_date DATE,
    period_month TINYINT,
    period_year YEAR,
    status ENUM('draft','issued','partial','paid') DEFAULT 'draft',
    issued_at DATETIME,
    paid_at DATETIME,
    created_by INT,
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (student_id) REFERENCES students(id)
);

CREATE TABLE payments (
    id INT PRIMARY KEY AUTO_INCREMENT,
    billing_id INT NOT NULL,
    payment_date DATE,
    amount DECIMAL(12,2) NOT NULL,
    payment_method ENUM('transfer','cash','qris','lainnya'),
    proof_file VARCHAR(500),
    notes TEXT,
    status ENUM('pending','verified','approved','rejected') DEFAULT 'pending',
    verified_by INT,
    verified_at DATETIME,
    receipt_number VARCHAR(30),
    created_at DATETIME DEFAULT NOW(),
    FOREIGN KEY (billing_id) REFERENCES billings(id)
);

-- =====
-- EVENTS
-- =====

CREATE TABLE events (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(200) NOT NULL,
    event_type ENUM('market_day','outbound','outing_class','pentas_seni',
                   'wisuda','parenting','camping','lainnya'),
    description TEXT,
    event_date DATE,
    location VARCHAR(200),
    max_participants INT,
    status ENUM('draft','published','registration_open',
               'ongoing','completed','cancelled') DEFAULT 'draft',
    created_by INT,
    created_at DATETIME DEFAULT NOW()
);

CREATE TABLE event_registrations (

```

```
id INT PRIMARY KEY AUTO_INCREMENT,  
event_id INT NOT NULL,  
participant_type ENUM('student','parent','teacher','external'),  
participant_id INT,  
participant_name VARCHAR(150),  
status ENUM('registered','attended','absent') DEFAULT 'registered',  
registered_at DATETIME DEFAULT NOW(),  
FOREIGN KEY (event_id) REFERENCES events(id)  
);
```

LEVEL 6 — APPLICATION ARCHITECTURE

6.1 Navigation Model per Stakeholder Type

ADMINISTRATOR (Kepala Sekolah / Admin)

Dashboard (Executive/Admin)

Master Data

← Administrator only

- └ Siswa
 - └ Daftar Siswa
 - └ Tambah Siswa
 - └ Import Siswa (CSV)
- └ Orang Tua / Wali Murid
 - └ Daftar Orang Tua
 - └ Hubungkan ke Siswa
- └ Guru / Fasilitator
 - └ Daftar Guru
 - └ Tambah Guru
- └ Kelas
 - └ Daftar Kelas
 - └ Penempatan Siswa
- └ Tahun Ajaran
 - └ Daftar Tahun Ajaran
 - └ Atur Tahun Aktif
- └ Mata Pelajaran / Tema
 - └ Daftar Mapel / Tema
 - └ Tambah Mapel / Tema
- └ Learning Outcome
 - └ Daftar Learning Outcome
 - └ Tambah / Edit Outcome

PPDB

- └ Calon Siswa
- └ Seleksi & Wawancara
 - └ Registrasi

Akademik

- └ Jadwal
- └ Manajemen Kelas
 - └ Penilaian

LOA Dashboard

Event

Finance

- └ Tagihan
 - └ Pembayaran

Laporan

- └ Laporan Akademik
- └ Laporan LOA
- └ Laporan Keuangan
 - └ Laporan Event

Sistem

- └ Pengguna
- └ Peran & Akses

- └ Audit Trail
- └ Konfigurasi

GURU / FASILITATOR (Operational User)

- Dashboard (Operational)
- Learning Planning
 - └ Semester Plan
 - └ Weekly Plan
 - └ Daily Plan [★ fitur utama – wizard form]
- Learning Execution
 - └ Absensi
 - └ Daily Activity
 - └ Observasi LOA
 - └ Daily Assessment
- Daily Report
 - └ Buat Laporan
 - └ Arsip Laporan
- Portfolio
 - └ Evidence & Portfolio
- Laporan Saya
 - └ Rekap Mingguan
 - └ Rekap Bulanan

WALI KELAS (Operational User)

- Dashboard (Kelas Saya)
- Kelas Saya
 - └ Daftar Siswa
 - └ Absensi Kelas
 - └ Progress Siswa
- Daily Report Kelas
- LOA Siswa
- Portfolio Siswa
- Rapor
 - └ Semester Assessment
 - └ Generate Rapor

ORANG TUA (External Party)

Beranda (Profil Anak)
Daily Report (harian dari guru)
Perkembangan Anak
└─ LOA Anak
└─ Portfolio Anak
Rapor Anak
Keuangan
└─ Tagihan & Pembayaran
Event Sekolah

SISWA (Internal User)

Beranda (Dashboard Siswa)
Jadwalku
Tugasku
Materi Belajar
Portofoliomu
Nilai & LOA
Rapor
Event Sekolah

STAFF KEUANGAN (Operational User)

Dashboard Keuangan
Tagihan
└─ Generate Tagihan
└─ Daftar Tagihan
└─ Rekap per Kelas
Pembayaran
└─ Verifikasi Pembayaran
└─ Riwayat Transaksi
Laporan Keuangan
└─ Koleksi SPP
└─ Rekapitulasi

PANITIA EVENT (Operational User)

Dashboard Event

- Event
 - Daftar Event
 - Buat Event Baru
 - Registrasi Peserta
- Pelaksanaan
 - Absensi Peserta
 - Rundown Acara
- Evaluasi & Laporan

AUDITOR (Support)

Dashboard Audit

- Audit Trail
 - Activity Log
 - Login History
 - Data Changes
- Kepatuhan
 - Business Rules Check
 - Compliance Report
- Audit Keuangan
- Audit Akademik

6.2 Screen Classification

Halaman	Pattern	Role
Login	Auth + Role Selector	Semua
Dashboard per role	Dashboard (role-based)	Per role
Daily Plan (wizard)	Wizard / Multi-step Form	Guru, Wali Kelas
List Siswa	CRUD List	Admin, Guru, Wali Kelas
Profil Siswa	Master Detail	Admin, Guru, Orang Tua
Observasi LOA	Form + Upload	Guru, Wali Kelas
Daily Report	Rich Form	Guru, Wali Kelas
Portfolio Siswa	Gallery / Timeline	Semua
Rapor	Master Detail + Print	Admin, Wali Kelas, Ortu
Tagihan	CRUD + Workflow	Staff Keuangan, Ortu

Halaman	Pattern	Role
Audit Trail	Timeline / Log	Auditor
Event Kalender	Calendar Board	Semua

6.3 Dashboard Specification per Role

Role	Widget Utama
Kepala Sekolah	Total siswa, Kehadiran hari ini, LOA achievement, Revenue bulan ini, Event mendatang, Monitoring alert
Admin	PPDB pending, Siswa baru, Tagihan overdue, Event aktif, Log aktivitas terkini
Guru/Fasilitator	Kelas hari ini, Absensi status, Aktivitas harian, Observasi terbaru, Daily plan status
Wali Kelas	Siswa hadir/absen kelas, LOA capaian kelas, Daily report status, Progress siswa
Siswa	Jadwal hari ini, Tugas aktif, LOA capaian, Portofolio terbaru
Orang Tua	Daily report terbaru, LOA anak, Tagihan status, Event mendatang
Staff Keuangan	Tagihan bulan ini, Terkumpul vs Target, Pembayaran pending, Overdue
Panitia Event	Event mendatang, Peserta terdaftar, Checklist persiapan
Auditor	Log hari ini, Anomali, Login aktif, Perubahan data

LEVEL 7 — IMPLEMENTATION ARCHITECTURE

7.1 Technology Stack

Layer	Technology
Frontend	Bootstrap 5.3 + Vanilla JS + jQuery 3.7
Backend	PHP 8.2 (Native, no framework)
Database	MySQL 8.0
Web Server	Apache 2.4 + mod_rewrite
Icons	Tabler Icons (CDN)

Layer	Technology
Font	Google Fonts (DM Sans + Lora)
Charts	Chart.js 4 (CDN)
File Upload	Native PHP + move_uploaded_file()
Session	PHP Native Session

7.2 Authentication Specification

Item	Value
Login Identifier	username
Credential	password (password_hash / PASSWORD_DEFAULT)
Session Name	SEKOLAHALAM_SESSION
Session Timeout	120 menit inaktif
Login Mode	Pill selector (demo) + form username/password
Role Source	Tabel users JOIN roles
Concurrent Login	Diizinkan
Logout	session_destroy()

Pill Selector Behavior:

- Halaman login menampilkan role-pill grid (sesuai data dari tabel roles)
- Saat user klik pill → username/password demo terisi otomatis (untuk mockup/demo)
- User tetap bisa isi manual username+password tanpa klik pill
- Setelah login berhasil → redirect ke dashboard-[role].php

7.3 Authorization Model

Stakeholder Type	Access Scope
Sponsor	Read-only executive dashboard & laporan
Decision Maker	Monitoring + approval + laporan domain
Operational User	Transaksi harian sesuai domain kerjanya
Administrator	Full access semua menu + system config

Stakeholder Type	Access Scope
Internal User	Portal siswa (read-only sebagian besar)
External Party	Portal orang tua (terbatas pada data anak sendiri)
Support	Audit trail + monitoring + compliance

7.4 Security Specification

Area	Strategy
Login	Username + password hash + CSRF token
Session	PHP native + timeout check per request
Page Guard	<code>auth_guard.php</code> di-include setiap halaman
Database	PDO + prepared statement (no raw query)
File Upload	Whitelist MIME, rename file, store outside webroot
Audit	Semua write operation log ke <code>audit_logs</code>

7.5 Coding Standard

Item	Standard
PHP Style	Native PHP 8.2
Database	PDO + prepared statement
Naming	snake_case (table, column, variable)
File	One purpose one file
Template	header.php + sidebar.php + footer.php
Session Variable	<code>\$_SESSION['user_id']</code> , <code>\$_SESSION['role']</code> , <code>\$_SESSION['full_name']</code>

LEVEL 8 — DEVELOPMENT ARCHITECTURE

8.1 Folder Structure

erp-sekolah-alam/

- |
- |— index.php ← redirect ke login
- |— login.php ← halaman login (pill selector + form)
- |— logout.php
- |— database.sql ← schema + seed data
- |
- |— config/
 - | |— database.php ← koneksi PDO
 - | |— auth.php ← auth helper (login, logout, check session)
 - | |— constants.php ← konstanta aplikasi
- |
- |— includes/
 - | |— auth_guard.php ← proteksi halaman (wajib include)
 - | |— header.php ← HTML head + meta
 - | |— topbar.php ← navbar atas
 - | |— sidebar-admin.php ← sidebar Administrator (3-level accordion)
 - | |— sidebar-guru.php ← sidebar Guru/Fasilitator (2-level treeview)
 - | |— sidebar-walikelas.php ← sidebar Wali Kelas (2-level treeview)
 - | |— sidebar-ortu.php ← sidebar Orang Tua (flat/1-level)
 - | |— sidebar-siswa.php ← sidebar Siswa (flat/1-level)
 - | |— sidebar-keuangan.php ← sidebar Staff Keuangan (2-level)
 - | |— sidebar-panitia.php ← sidebar Panitia Event (flat/2-level)
 - | |— sidebar-auditor.php ← sidebar Auditor (2-level)
 - | |— sidebar-kepala.php ← sidebar Kepala Sekolah (2-level)
 - | |— footer.php ← HTML closing
 - | |— functions.php ← helper functions global
- |
- |— assets/
 - | |— css/
 - | | |— theme.css ← global CSS variables + layout
 - | | |— sidebar.css ← sidebar styles
 - | | |— components.css ← card, badge, button, form styles
 - | |— js/
 - | | |— sidebar.js ← accordion/treeview behavior
 - | | |— app.js ← global JS helpers
 - | | |— daily-plan-wizard.js ← wizard form logic
 - | |— uploads/ ← file upload storage
 - | | |— evidence/
 - | | |— portfolio/
 - | | |— payment-proof/
 - | | |— avatars/
- |
- |— pages/
 - | |— dashboard/
 - | | |— kepala-sekolah.php

- └─ admin.php
- └─ guru.php
- └─ walikelas.php
- └─ siswa.php
- └─ ortu.php
- └─ keuangan.php
- └─ panitia.php
- └─ auditor.php

- └─ learning-planning/

- └─ semester-plan/

- └─ index.php ← list
 - └─ form.php ← create/edit
 - └─ view.php ← detail

- └─ weekly-plan/

- └─ index.php
 - └─ form.php
 - └─ view.php

- └─ daily-plan/

- └─ index.php ← list daily plan
 - └─ wizard.php ← WIZARD FORM (★ fitur utama)
 - └─ view.php ← detail plan
 - └─ save.php ← handler POST

- └─ daily-execution/

- └─ attendance/

- └─ index.php
 - └─ save.php

- └─ observation/

- └─ index.php
 - └─ form.php
 - └─ save.php
 - └─ upload-evidence.php

- └─ daily-report/

- └─ index.php
 - └─ form.php
 - └─ save.php
 - └─ view.php ← tampilan untuk orang tua

- └─ portfolio/

- └─ index.php ← gallery portofolio

- └─ item/

- └─ form.php
 - └─ save.php

- └─ student/

- └─ view.php ← portofolio 1 siswa

Domain	Module	File Utama
Portfolio	Evidence + Gallery	pages/portfolio/
Students	CRUD Siswa	pages/students/
PPDB	Workflow Admisi	pages/ppdb/
LOA	Definisi + Laporan	pages/loa/
Finance	Billing + Payment	pages/finance/
Events	Event + Registrasi	pages/events/
System	Users + Audit	pages/system/

8.3 Sprint Planning

Sprint	Scope	Estimasi
Sprint 1	Auth (login + pill selector) + Sidebar Navigation (9 role) + Dashboard (9 role)	Done first
Sprint 2	Learning Planning (Semester, Weekly, Daily Plan Wizard)	Setelah S1 approved
Sprint 3	Daily Execution (Absensi, Observasi LOA, Daily Report)	Setelah S2
Sprint 4	Portfolio + Progress + Semester Report	Setelah S3
Sprint 5	PPDB + Students + Parents CRUD	Paralel S2-S3
Sprint 6	Finance (Billing + Payment)	Setelah S5
Sprint 7	Events + System (Audit, Users)	Setelah S6
Sprint 8	Reports + Print + Refinement	Finalisasi

LEVEL 9 — CODE GENERATION ARCHITECTURE

9.1 Source Code Generation Convention

Item	Convention
Application Style	Native PHP 8.2, no framework
Database Access	PDO + prepared statement
Frontend	Bootstrap 5.3 + Tabler Icons + Chart.js
JavaScript	Vanilla JS + jQuery 3.7
Naming	snake_case (DB), PascalCase (tidak dipakai)
File Organization	One file one purpose
Navigation	Multi-file, <code><a href></code>
Template	header.php + sidebar.php + footer.php include
Session	PHP Native <code>\$_SESSION</code>
CRUD Convention	index.php, form.php, save.php, view.php, delete.php

9.2 Authentication Flow Code Pattern

php

```

// login.php
// 1. GET: tampilkan form + pill selector (load roles dari DB)
// 2. POST: validasi → cek users table → set session → redirect

// auth_guard.php (di-include setiap halaman)
session_start();
if (!isset($_SESSION['user_id'])) {
    header('Location: /login.php');
    exit;
}
// Check timeout
if (time() - $_SESSION['last_activity'] > 7200) {
    session_destroy();
    header('Location: /login.php?timeout=1');
    exit;
}
$_SESSION['last_activity'] = time();

// Redirect after login berdasarkan role
$role_redirect = [
    'kepala_sekolah' => '/pages/dashboard/kepala-sekolah.php',
    'admin'          => '/pages/dashboard/admin.php',
    'guru'           => '/pages/dashboard/guru.php',
    'walikelas'      => '/pages/dashboard/walikelas.php',
    'siswa'          => '/pages/dashboard/siswa.php',
    'ortu'           => '/pages/dashboard/ortu.php',
    'keuangan'       => '/pages/dashboard/keuangan.php',
    'panitia'        => '/pages/dashboard/panitia.php',
    'auditor'        => '/pages/dashboard/auditor.php',
];

```

9.3 Sidebar Pattern per Role

```

Administrator → sidebar-admin.php → Accordion 3-level (Domain→Group→Item)
Kepala Sekolah → sidebar-kepala.php → Treeview 2-level (Section→Item)
Guru           → sidebar-guru.php → Treeview 2-level (Section→Item)
Wali Kelas     → sidebar-walikelas.php → Treeview 2-level
Siswa         → sidebar-siswa.php → Flat 1-level (semua menu langsung)
Orang Tua      → sidebar-ortu.php → Flat 1-level (sederhana)
Staff Keuangan → sidebar-keuangan.php → Treeview 2-level
Panitia Event  → sidebar-panitia.php → Flat/2-level sederhana
Auditor        → sidebar-auditor.php → Treeview 2-level

```

9.4 Daily Plan Wizard — Step Structure

Step 1: Informasi Umum

- Kelas (dropdown), Tanggal, Periode, Minggu
- Fasilitator (auto dari session)

Step 2: Tema & Tujuan

- Tema Besar (master selector dari semester plan)
- Sub Tema / Judul Kegiatan Hari Ini
- Tujuan Pembelajaran (textarea)
- Fokus Karakter / Akhlak (multi-select checkbox)
- Referensi Ayat Al-Quran (textarea)

Step 3: Rencana Aktivitas

- Tabel dinamis aktivitas (add row / drag sort)
- Per baris: Waktu Mulai, Waktu Selesai, Judul Aktivitas, Jenis (outdoor/indoor/dll), Deskripsi, LOA target

Step 4: Materi & Penutup

- Materi/Alat yang dibutuhkan (list dinamis)
- Kegiatan Pembuka
- Kegiatan Penutup
- Catatan tambahan
- Preview & Submit / Save Draft

9.5 Database Seed Data (untuk Demo/Mockup)

sql

```
-- Roles (9 role sesuai stakeholder)
INSERT INTO roles VALUES
(1,'kepala_sekolah','Kepala Sekolah','decision_maker','Pimpinan sekolah'),
(2,'admin','Admin Sekolah','administrator','Operator sistem'),
(3,'guru','Guru / Fasilitator','operational_user','Mengajar & observasi LOA'),
(4,'walikelas','Wali Kelas','operational_user','Mengelola kelas & rapor'),
(5,'siswa','Siswa','internal_user','Peserta didik'),
(6,'ortu','Orang Tua / Wali Murid','external_party','Pemantau perkembangan anak'),
(7,'keuangan','Staff Keuangan','operational_user','Pengelola keuangan'),
(8,'panitia','Panitia Event','operational_user','Pengelola event sekolah'),
(9,'auditor','Auditor Internal','support','Pengawas kepatuhan');

-- Users demo (1 per role)
INSERT INTO users (username,password,full_name,role_id,is_active) VALUES
('kepala','[hash]','Dr. Hendra Wijaya',1,1),
('admin','[hash]','Ratna Sari, S.Pd',2,1),
('guru','[hash]','Ibu Sari Wulandari',3,1),
('walikelas','[hash]','Pak Bimo Susanto',4,1),
('siswa','[hash]','Arjuna Pratama',5,1),
('ortu','[hash]','Budi Pratama',6,1),
('keuangan','[hash]','Dewi Lestari, S.E',7,1),
('panitia','[hash]','Pak Rudi Santoso',8,1),
('auditor','[hash]','Hadi Nugroho, S.H',9,1);
```

KEPUTUSAN DESAIN PENTING

No	Topik	Keputusan
1	Visual Theme	Earth/Nature Light — cream (☐ #faf8f4) + forest green (☒ #1b4332)
2	Sidebar Admin	Accordion 3-level (Berlian style)
3	Sidebar Non-Admin	Flat atau Treeview 2-level (sesuai kompleksitas role)
4	Login Mode	Pill selector (role dari DB) + form manual (dua mode)
5	Daily Plan	Wizard 4-step, terinspirasi dari daily-plan-form.html
6	LOA Evidence	Wajib upload minimal 1 sebelum submit observasi
7	Daily Report	Dipublish ke semua orang tua dalam satu kelas

No	Topik	Keputusan
8	Rapor	Berbasis OBE (evidence + outcome + daily assessment)
9	Portfolio	Dibuat otomatis dari evidence yang diupload
10	Stack	Native PHP 8.2 + MySQL 8 + Bootstrap 5, NO framework

Dokumen ini adalah living document — revisi dilakukan sebelum Sprint coding dimulai.